

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Single Correct

Match List I with List II.

List I

(Monomer Unit)

(a) Caprolactum

(b) 2-Chloro-1,3-butadiene

(c) Isoprene

(d) Acrylonitrile

List II

(Polymer)

(i) Natural rubber

(ii) Buna-N

(iii) Nylon 6

(iv) Neoprene

Choose the correct answer from the options given below :

(1) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (i), (d) $\rightarrow$ (ii)(2) (a) $\rightarrow$ (i), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (iv)(3) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (iii)(4) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)

Q2: 24 Feb (Shift 2) - Single Correct

In polymer Buna-S; 'S' stands for :

(1) Styrene

(2) Sulphur

(3) Strength

(4) Sulphonation

Q3: 25 Feb (Shift 1) - Single Correct

Which statement is correct?

(1) Buna-S is a synthetic and linear thermosetting polymer

(2) Neoprene is addition copolymer used in plastic bucket manufacturing

(3) Synthesis of Buna-S needs nascent oxygen

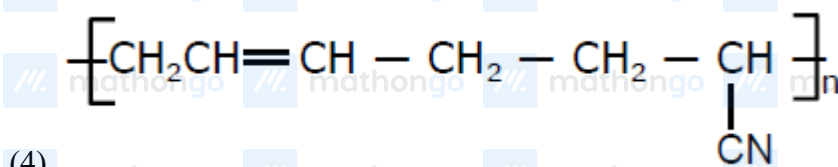
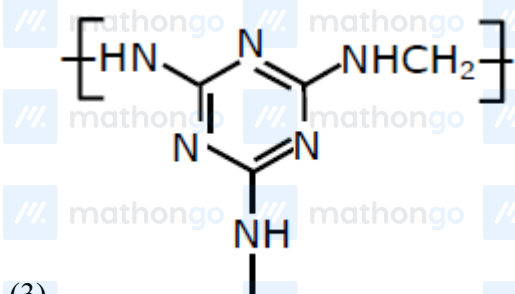
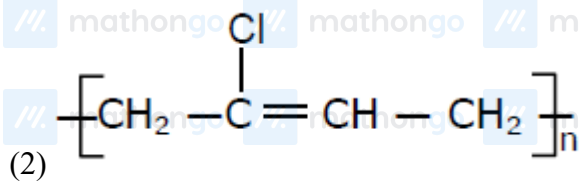
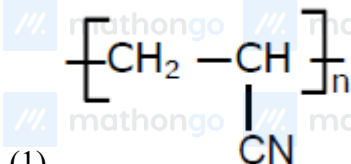
Questions with Answer Keys

MathonGo

(4) Buna-N is a natural polymer

Q4: 26 Feb (Shift 1) - Single Correct

The structure of Neoprene is :



Answer Key**Q1 (1)****Q2 (1)****Q3 (3)****Q4 (2)**

Q1: 24 Feb (Shift 1) - Single Correct

Out of the following, which type of interaction is responsible for the stabilisation α -helix structure of proteins?

- (1) Ionic bonding
- (2) Hydrogen bonding
- (3) vander Waals forces
- (4) Covalent bonding

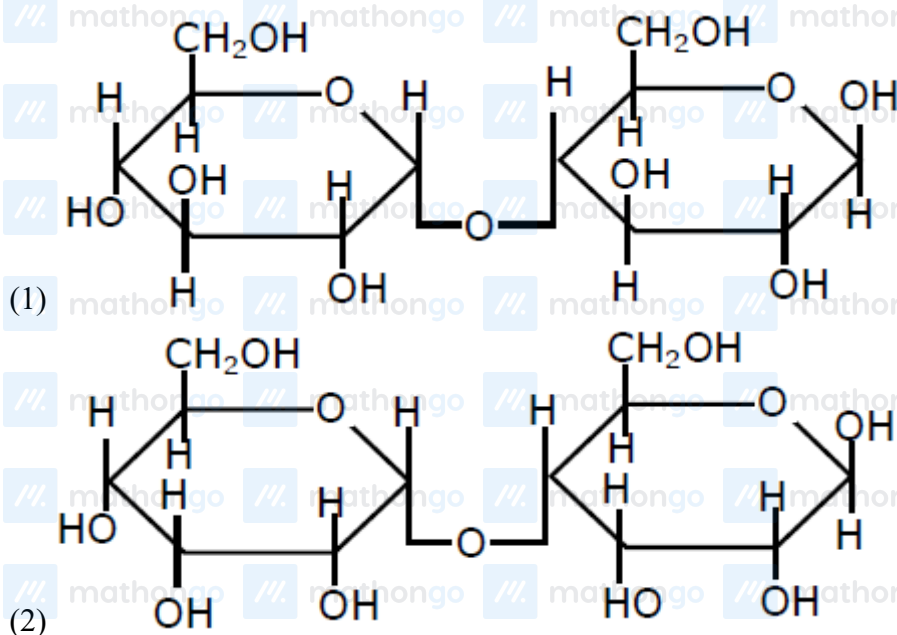
Q2: 25 Feb (Shift 1) - Single Correct

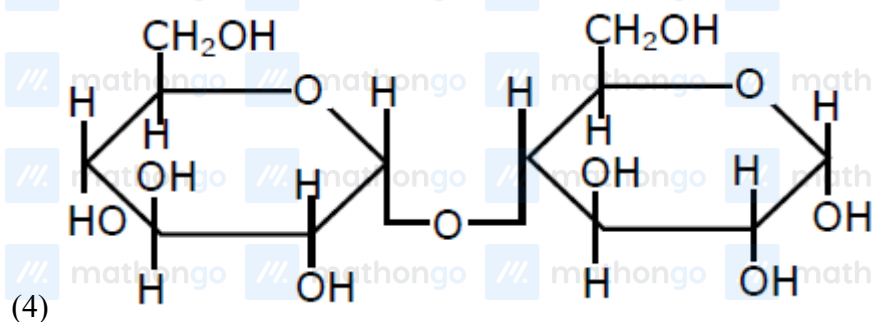
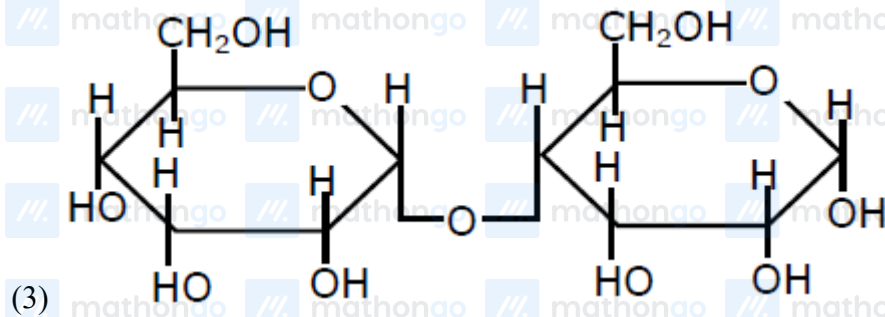
Which of the glycosidic linkage galactose and glucose is present in lactose?

- (1) C – 1 of glucose and C – 6 of galactose
- (2) C – 1 of galactose and C – 4 of glucose
- (3) C – 1 of glucose and C – 4 of galactose
- (4) C – 1 of galactose and C – 6 of glucose

Q3: 25 Feb (Shift 2) - Single Correct

Which of the following is correct structure of α -anomer of maltose :





Q4: 26 Feb (Shift 1) - Single Correct

Which of the following vitamin is helpful in delaying the blood clotting ?

- (1) Vitamin B
- (2) Vitamin C
- (3) Vitamin K
- (4) Vitamin E

Q5: 26 Feb (Shift 2) - Single Correct

Match List-I with List-II.

List-I

- (a) Sucrose
- (b) Lactose
- (c) Maltose

List-II

- (i) β -D-Galactose and β -D-Glucose
- (ii) α -D-Glucose and β -D-Fructose
- (iii) α -D-Glucose and α -D-Glucose

Choose the correct answer from the options given below:

- (1) (a) – (iii), (b) – (ii), (c) – (i)

Questions with Answer Keys

MathonGo

(2) (a) – (iii), (b) – (i), (c) – (ii)

(3) (a) – (i), (b) – (iii), (c) – (ii)

(4) (a) – (ii), (b) – (i), (c) – (iii)

Q6: 26 Feb (Shift 2) - Single Correct

Seliwanoff test and Xanthoproteic test are used for the identification of _____ and _____ respectively

(1) ketoses, proteins

(2) proteins, ketoses

(3) aldoses, ketoses

(4) ketoses, aldoses

Answer Key**Q1 (2)****Q2 (2)****Q3 (4)****Q4 (3)****Q5 (4)****Q6 (1)**

Hints and Solutions

MathonGo

Q1 (1)

- (1) Polymer of caprolactum is nylon-6
- (2) Polymer of 2-chloro-1,3-butadiene is neoprene.
- (3) Polymer of isoprene is natural rubber
- (4) Polymer of acrylonitrile and 1,3-butadiene is buna-N

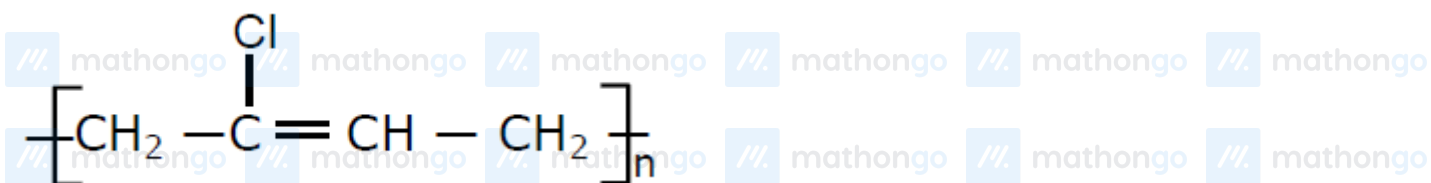
Q2 (1)

Buna-S is the co-polymer of buta-1,3 diene & styrene.

Q3 (3)

Synthesis of Buna-S needs nascent oxygen.

Q4 (2)

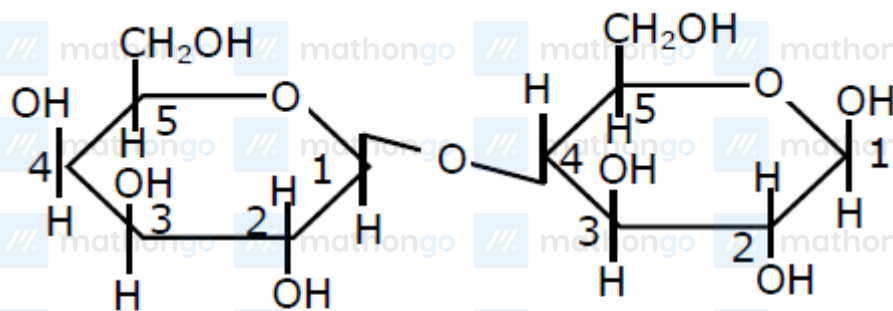


neoprene

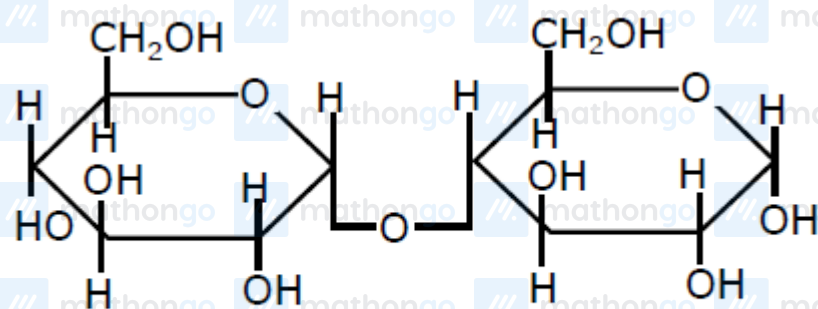
Q1 (2)

The α -helix is stabilized by hydrogen bond between the NH and CO group of the main chain.

Q2 (2)

 β -D-Galactose β -D-Glucose

Q3 (4)

[α -Anomer of maltose]

Q4 (3)

Vitamin K is used by the body to help blood clot.

Q5 (4)

Sucrose $\rightarrow \alpha$ - D - Glucose and β - D - Fructose

Lactose $\rightarrow \beta$ - D - Galactose and β - D - Glucose

Maltose $\rightarrow \alpha$ -D- Glucose and α -D- Glucose

Q6 (1)

Seliwanoff test and Xanthoproteic test are used for identification of 'Ketoses' and proteins respectively.